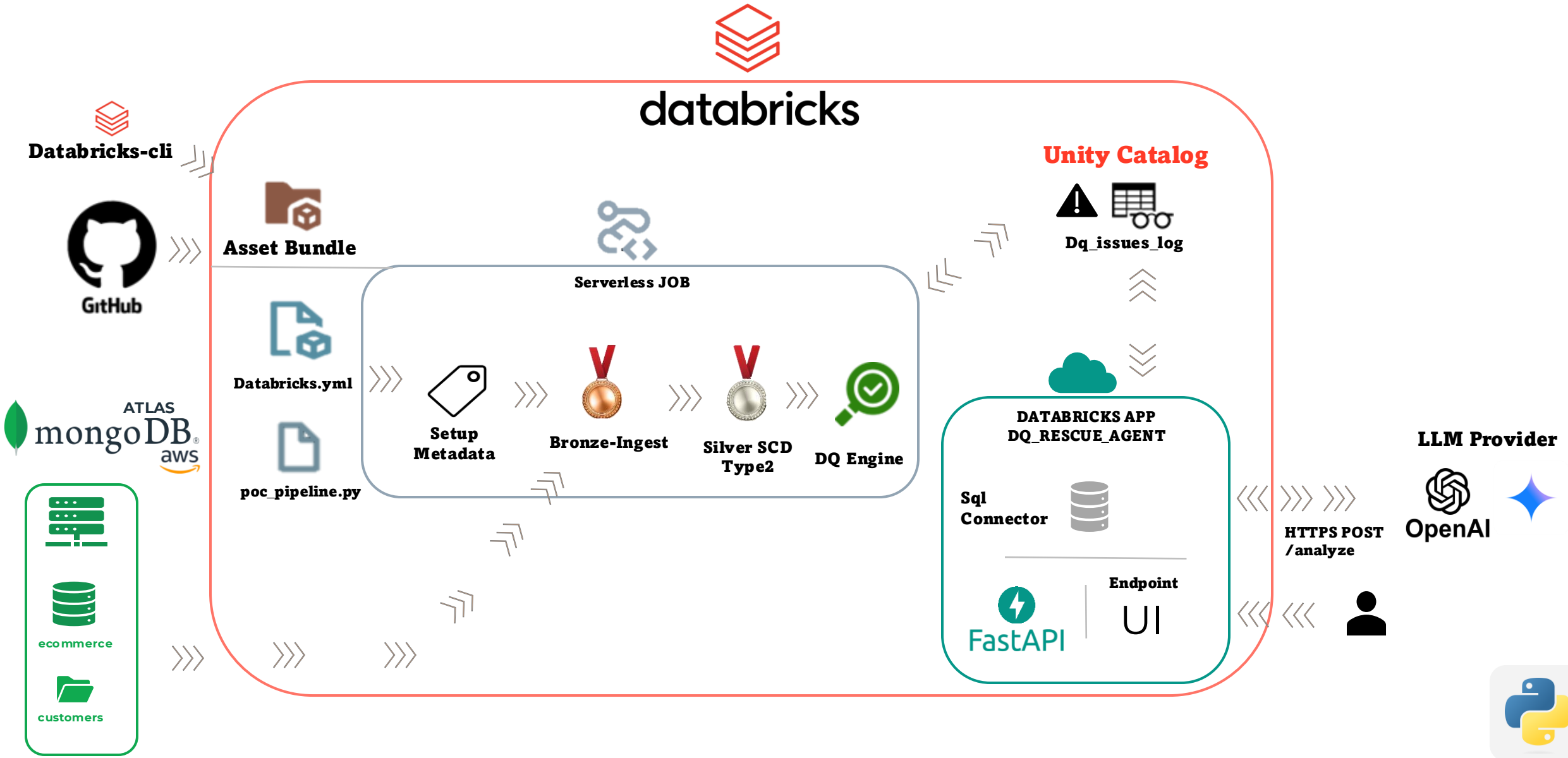
Metadata-driven Agentic DQ in Databricks, Serverless, FastAPI + Gemini, MongoDB



File Edit Selection View Go Run Terminal Help

Databrics-MetadataDriven-SCDT2-Scala-AgenticDataQApp

EXPLORER

DATABRICS-METADATADRIVEN-SCDT2-SCALA-AGENTICDATAQAPP

- .databricks
- .metals
- .venv
- .vscode
- app
- docs
  - AGENTIC_DATA_QUALITY_POC_TECHNICAL_DOCUMENTATION.md
  - AGENTIC_DATA_QUALITY_POC_TECHNICAL_DOCUMENTATION.pdf
  - AGENTIC_DATA_QUALITY_POC_TECHNICAL_DOCUMENTATION.txt
- pipeline
  - __pycache__
  - poc_pipeline.py
- scripts
  - __pycache__
  - 00_mongo_simulator.py
  - text_to_pdf.py
- target
- .env
- .env.example
- .gitignore
- .load-env.ps1
- commands_guide.txt
- databricks.yml
- README.md

OUTLINE

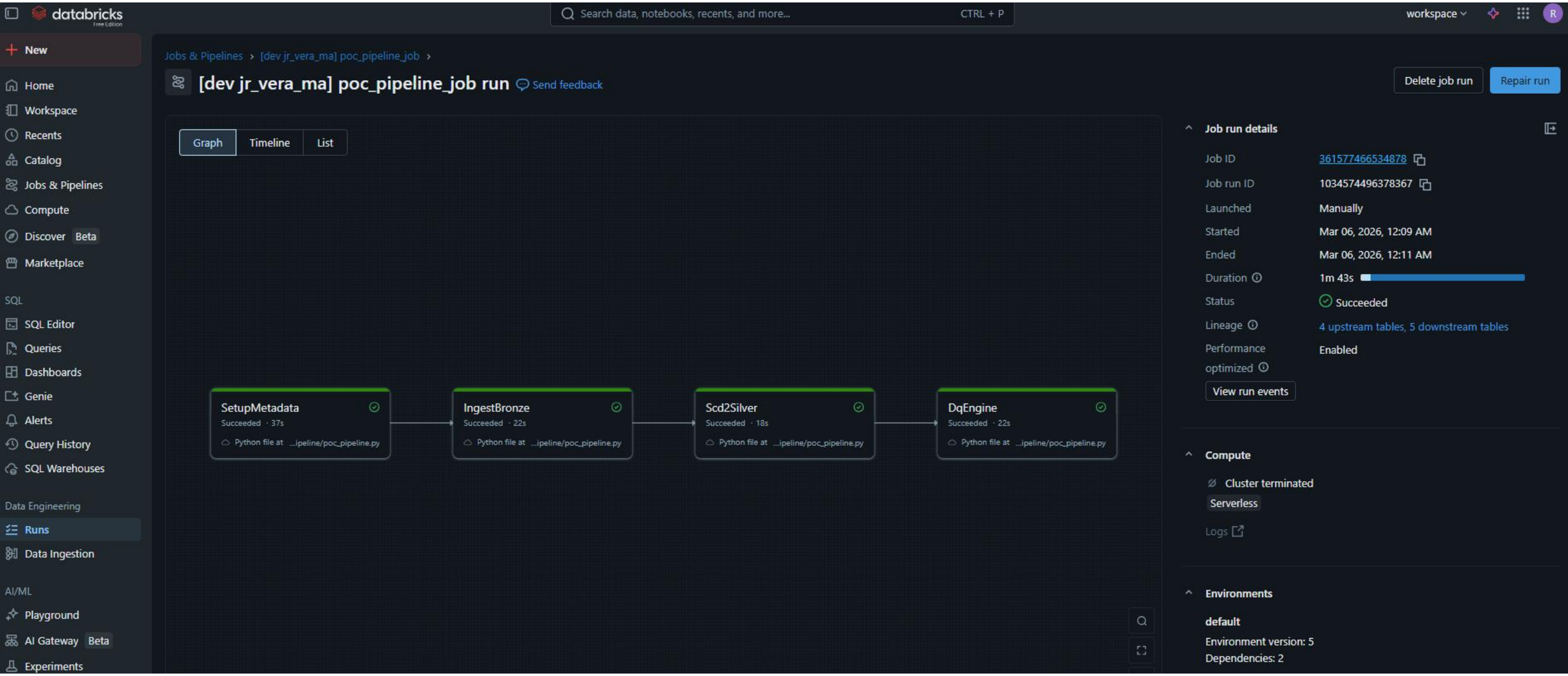
TIMELINE

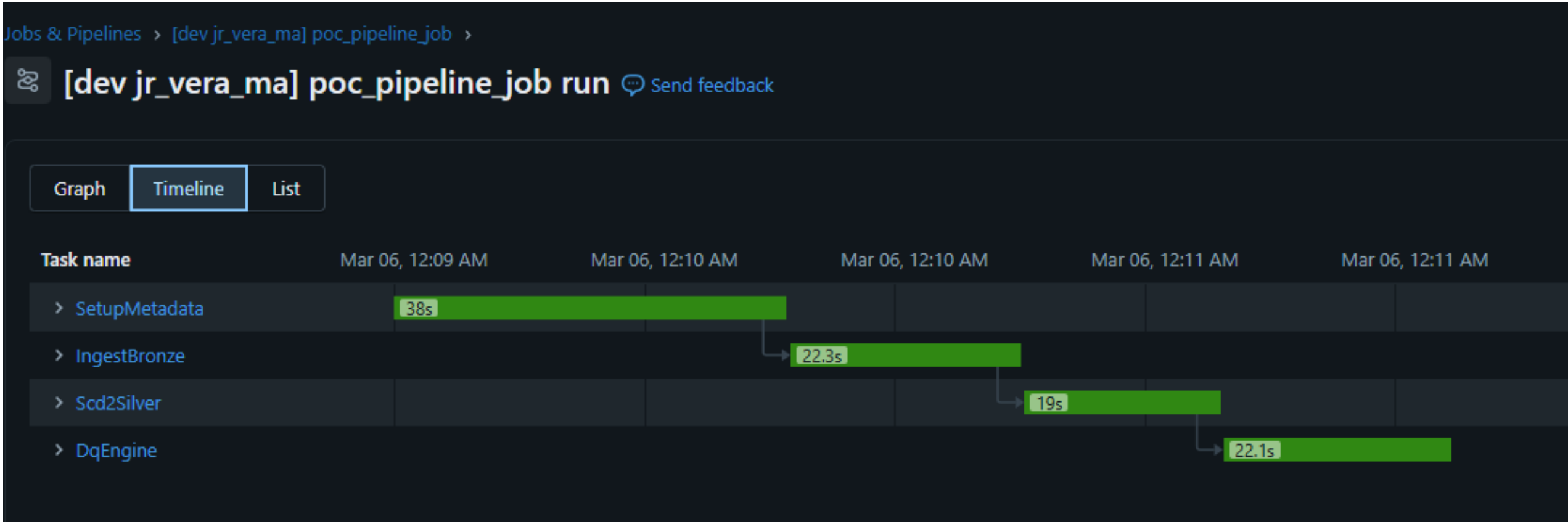
commands_guide.txt

```
databricks bundle run -t dev dq_rescue_agent
# Open the app URL in browser:
# - UX interface: https://<your-app-url>/ui
# - API docs:      https://<your-app-url>/docs
# Optional API checks (PowerShell):
# Invoke-RestMethod -Method Get -Uri "https://<your-app-url>/issues-summary?window_hours=24&max_issues=200&sample_size=3"
# Invoke-RestMethod -Method Post -Uri "https://<your-app-url>/analyze" -ContentType "application/json" -Body '{"analysis_mode":"summ
# Invoke-RestMethod -Method Post -Uri "https://<your-app-url>/analyze" -ContentType "application/json" -Body '{"analysis_mode":"late

## 8) Simulate incremental good and bad records
.\.venv\Scripts\Activate.ps1
python scripts/00_mongo_simulator.py --mode add_good
python scripts/00_mongo_simulator.py --mode add_bad
# PowerShell one-liners (loads env + runs):
. .\.load-env.ps1; python scripts/00_mongo_simulator.py --mode add_good
. .\.load-env.ps1; python scripts/00_mongo_simulator.py --mode add_bad

## 9) Rerun the workflow after new data
databricks bundle run -t dev poc_pipeline_job

[ DATABRICKS APP URL : Paste link here after deployment ]
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

```
(.venv) PS C:\Users\jrver\PycharmProjects\Python\Databrics-MetadataDriven-SCDT2-Scala-AgenticDataQApp> databricks bundle validate -t dev
Target: dev
Workspace:
  Host: https://dbc-78d92c1b-3cff.cloud.databricks.com
  User: jr.vera.ma@gmail.com
  Path: /Workspace/Users/jr.vera.ma@gmail.com/.bundle/agentic_dq_poc/dev

Validation OK!
(.venv) PS C:\Users\jrver\PycharmProjects\Python\Databrics-MetadataDriven-SCDT2-Scala-AgenticDataQApp> databricks bundle deploy -t dev
Uploading bundle files to /Workspace/Users/jr.vera.ma@gmail.com/.bundle/agentic_dq_poc/dev/files...
Deploying resources...
Updating deployment state...
Deployment complete!
(.venv) PS C:\Users\jrver\PycharmProjects\Python\Databrics-MetadataDriven-SCDT2-Scala-AgenticDataQApp> databricks bundle run -t dev dq_rescue_agent
✓ Getting the status of the app dq-rescue-agent
✓ App is in RUNNING state
✓ App compute is in ACTIVE state
✓ Preparing source code for new app deployment.
✓ Deployment is pending
✓ Starting app with command: uvicorn agent_app:app --host 0.0.0.0 --port 8000
✓ App started successfully
You can access the app at https://dq-rescue-agent-1389190390101071.aws.databricksapps.com
(.venv) PS C:\Users\jrver\PycharmProjects\Python\Databrics-MetadataDriven-SCDT2-Scala-AgenticDataQApp>
```


databricks
Free Edition
Search data, notebooks, recents, and more...
CTRL + P
workspace
New
Home
Workspace
Recents
Catalog
Jobs & Pipelines
Compute
Discover Beta
Marketplace
SQL
SQL Editor
Queries
Dashboards
Genie
Alerts
Query History
SQL Warehouses
Data Engineering
Runs
Data Ingestion
AI/ML
Playground
AI Gateway Beta
Experiments
Jobs & Pipelines › [dev jr_vera_ma] poc_pipeline_job ›
[dev jr_vera_ma] poc_pipeline_job run
Send feedback
Delete job run
Repair run
Graph
Timeline
List
SetupMetadata
Succeeded · 37s
Python file at ...ipeline/poc_pipeline.py
IngestBronze
Succeeded · 22s
Python file at ...ipeline/poc_pipeline.py
Scd2Silver
Succeeded · 18s
Python file at ...ipeline/poc_pipeline.py
DqEngine
Succeeded · 22s
Python file at ...ipeline/poc_pipeline.py
Job run details
Job ID 361577466534878
Job run ID 1034574496378367
Launched Manually
Started Mar 06, 2026, 12:09 AM
Ended Mar 06, 2026, 12:11 AM
Duration 1m 43s
Status Succeeded
Lineage 4 upstream tables, 5 downstream tables
Performance optimized Enabled
View run events
Compute
Cluster terminated
Serverless
Logs
Environments
default
Environment version: 5
Dependencies: 2


Jobs & Pipelines › [dev jr_vera_ma] poc_pipeline_job ›
[dev jr_vera_ma] poc_pipeline_job run
Send feedback
Graph
Timeline
List
Task name
Mar 06, 12:09 AM
Mar 06, 12:10 AM
Mar 06, 12:10 AM
Mar 06, 12:11 AM
Mar 06, 12:11 AM
SetupMetadata
38s
IngestBronze
22.3s
Scd2Silver
19s
DqEngine
22.1s

Jobs & Pipelines › [dev jr_vera_ma] poc_pipeline_job ›

[dev jr_vera_ma] poc_pipeline_job run

Send feedback

Graph | Timeline | List

Filter task names starting with...

Status | Type

Status	Name	Type	Resource	Duration	Depends on
Succeeded	SetupMetadata	Python script	/Workspace/Users/jr.vera.m	37s	
Succeeded	IngestBronze	Python script	/Workspace/Users/jr.vera.m	22s	SetupMetadata
Succeeded	Scd2Silver	Python script	/Workspace/Users/jr.vera.m	18s	IngestBronze
Succeeded	DqEngine	Python script	/Workspace/Users/jr.vera.m	22s	Scd2Silver

Jobs & Pipelines ›

[dev jr_vera_ma] poc_pipeline_job

Send feedback

Run n

Connected to Databricks Asset Bundles Learn more

Open in Git folder

Disconnect fro

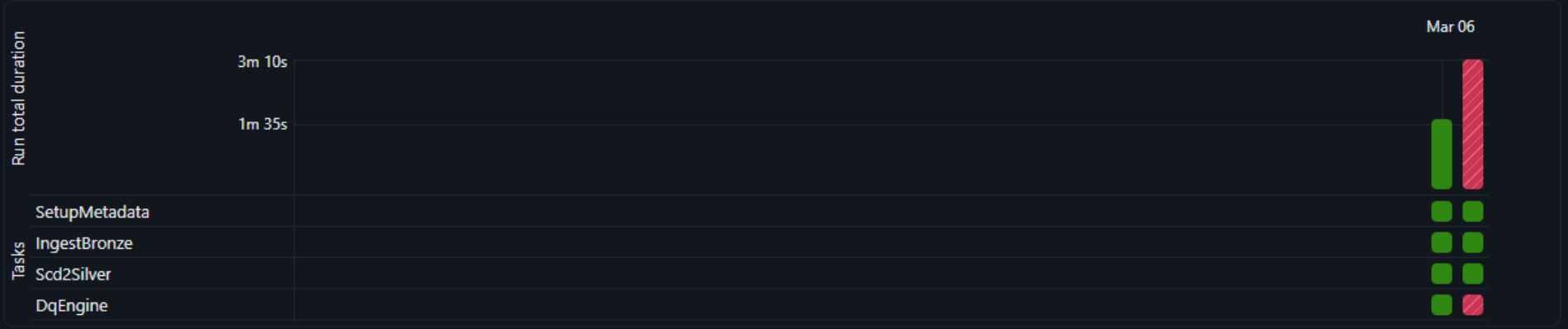
Runs Tasks

Runs

Started before

Previous

Next



Go to the latest successful run

Cancel runs

Start time	Run ID	Launched	Duration	Status	Error code	Run parameters
Mar 06, 2026, 12:47 AM	581318365317811	Manually	3m 11s	Failed	RunExecutionError	
Mar 06, 2026, 12:09 AM	1034574496378367	Manually	1m 43s	Succeeded		

Job details

Job ID	361577466534878
Creator	Rafa Vera
Run as	Rafa Vera
Description	Add description
Lineage	4 upstream tables, 5 downstream tables
Performance optimized	On

Schedules & Triggers

None

Add trigger

Job parameters

No job parameters are defined for this job

Edit parameters

Compute

Serverless

Jobs & Pipelines › [dev jr_vera_ma] poc_pipeline_job ›

[dev jr_vera_ma] poc_pipeline_job run

Send feedback

Delete job run | Repair

Graph | Timeline | List

Job run details

Job ID	361577466534878
Job run ID	581318365317811
Launched	Manually
Started	Mar 06, 2026, 12:47 AM
Ended	Mar 06, 2026, 12:51 AM
Duration	3m 10s
Status	Failed
Lineage	4 upstream tables, 5 downstream tables
Performance optimized	Enabled

View run events

Compute

Cluster terminated

Serverless

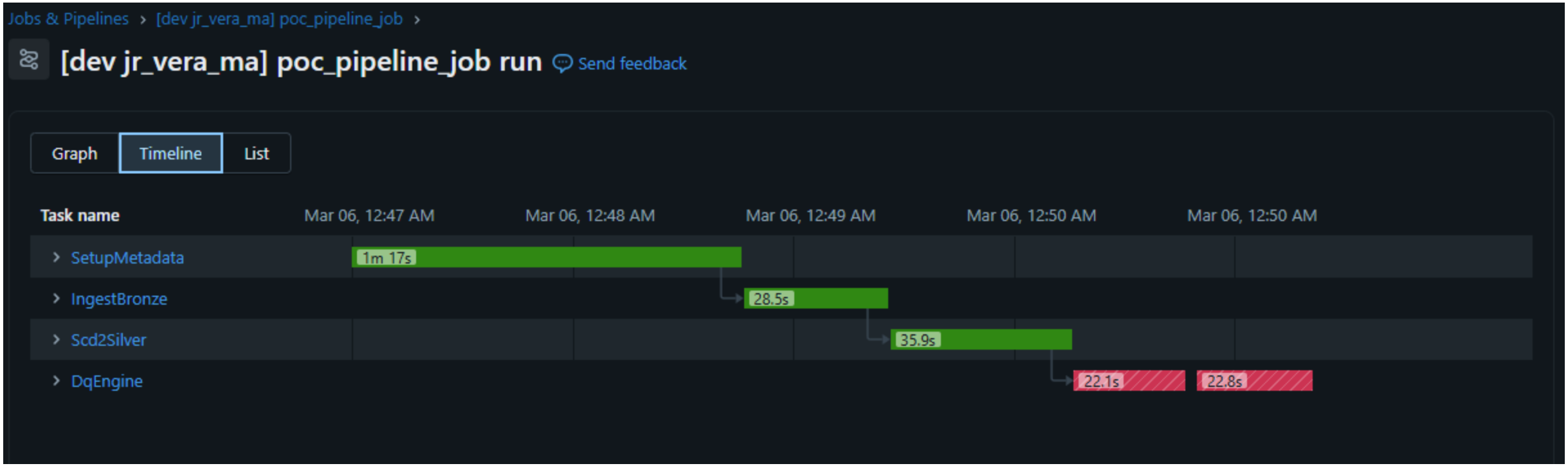
Logs

Environments

default

Environment version: 5

Dependencies: 2


Jobs & Pipelines › [dev jr_vera_ma] poc_pipeline_job ›
[dev jr_vera_ma] poc_pipeline_job run
Send feedback
Graph
Timeline
List
Task name
Mar 06, 12:47 AM
Mar 06, 12:48 AM
Mar 06, 12:49 AM
Mar 06, 12:50 AM
Mar 06, 12:50 AM
SetupMetadata
1m 17s
IngestBronze
28.5s
Scd2Silver
35.9s
DqEngine
22.1s
22.8s

Jobs & Pipelines › [dev jr_vera_ma] poc_pipeline_job › Run 581318365317811 ›

DqEngine run

Retry: 1st (latest) · Failed

Output

```
RuntimeError: Data quality checks failed with 2 issues
---------------------------------------------------------------------------
RuntimeError                              Traceback (most recent call last)
File ~/.ipykernel/16438/command--1-2794560781:9
      6 del sys
      8 with open(filename, "rb") as f:
----> 9   exec(compile(f.read(), filename, 'exec'))

File /Workspace/Users/jr.vera.ma@gmail.com/.bundle/agentic_dq_scala_poc/dev/files/pipeline/poc_pipeline.py:548
    544         spark.stop()
    547 if __name__ == "__main__":
--> 548     main()

File /Workspace/Users/jr.vera.ma@gmail.com/.bundle/agentic_dq_scala_poc/dev/files/pipeline/poc_pipeline.py:533, in main(
```

[dev jr_vera_ma] poc_pipeline_job run

Send feedback

Graph | Timeline | List

Filter task names starting with...

Status | Type

Status	Name	Type	Resource	Duration	Depends on
Succeeded	SetupMetadata	Python script	/Workspace/Users/jr.vera.m	1m 16s	
Succeeded	IngestBronze	Python script	/Workspace/Users/jr.vera.m	28s	SetupMetadata
Succeeded	Scd2Silver	Python script	/Workspace/Users/jr.vera.m	35s	IngestBronze
Failed	DqEngine	Python script	/Workspace/Users/jr.vera.m	22s	Scd2Silver

Search data, notebooks, recents, and more... CTRL + P

workspace

Compute › Apps ›

dq-rescue-agent

Send feedback | Edit | Permissions | De

Overview | Authorization | Deployments | Logs | Insights Beta | Environment

App status

Running https://dq-rescue-agent-1389190390101071.aws.databricksapps.com

Compute status: Active

Source

/Workspace/Users/jr.vera.ma@gmail.com/.bundle/agentic_dq_poc/dev/files/app

Deployment

Source: /Workspace/Users/jr.vera.ma@gmail.com/.bundle/agentic_dq_poc/dev/files/app

✓ Last deploy completed 35 seconds ago

Inspect deployed code

```
Mar 06, 2026, 02:24:50 PM GMT+1 ✓ Stopped app successfully
Mar 06, 2026, 02:24:51 PM GMT+1 ✓ Source code downloaded successfully
Mar 06, 2026, 02:24:51 PM GMT+1 ✓ App spec loaded successfully
Mar 06, 2026, 02:25:00 PM GMT+1 ✓ Packages installed successfully
Mar 06, 2026, 02:25:00 PM GMT+1 ✓ App built successfully
Mar 06, 2026, 02:25:02 PM GMT+1 ✓ App started successfully
```

App resources

Add

About this app

Creator	jr.vera.ma@gmail.com
Created on	15 hours ago
Last update	1 minute ago
App ID	1764d100-b946-4813-91

Compute

Medium (Up to 2 vCPUs, 6 GB memory, DBU/hour)

Tags

Add tags

dq-rescue-agent-1389190390101071.aws.databricksapps.com/ui

Capgemini | IA | Articulos | APIs | CloudPlatforms | BBDD | Empleo | Forecasting | GITHUB | Learning | My Web | NOISE PROJECT | MASTER | OPENAI | Pyth

Data Quality Agent

Interactive view on top of /latest-issue and /analyze

Open API docs | Health

Analysis completed (summary mode).

Issue context

Use latest issue or aggregated summary, then run concise technical analysis.

Load latest issue | Load summary | Analyze now | Clear

ANALYSIS MODE: summary

CONCISE OUTPUT: enabled

WINDOW HOURS: 24

MAX ISSUES: 200

SAMPLE SIZE: 3

RULE: email_not_null

SEVERITY: ERROR

RECORD KEY: C006

TIMESTAMP: 2026-03-05T23:50:59.018329Z

Additional context (optional)

Example: Pipeline run after add_bad failed at DqEngine with 2 issues

Issues summary payload


```
{
  "window_hours": 24,
  "total_issues": 4,
  "distinct_rules": 2,
  "distinct_records": 2,
  "rules": [
    {
      "rule_id": "R002",
      "rule_name": "email_not_null",
      "severity": "ERROR",
      "failed_condition": "email IS NOT NULL",
      "failed_rows": 2,
      "distinct_records": 1,
      "latest_timestamp": "2026-03-05T23:50:59.018329Z"
    },
    {
      "rule_id": "R001",
      "rule_name": "price_positive",
      "severity": "ERROR",
      "failed_condition": "price > 0",
      "failed_rows": 2,
```


Agent analysis

Direct, technical remediation based on selected analysis mode.

Analysis output

Here's an analysis of the data quality incident and proposed remediation:

---

1) Root cause

The incident reveals two distinct data quality issues originating from the MongoDB source, propagating through the pipeline to the `silver_customers` table, and subsequently flagged by the `DqEngine`.

- **Rule R002: `email_not_null` (for `customer_id: C006`)**
- **What failed**: The `email` field is entirely absent from the source record for `customer_id: C006`. When ingested into a structured format like Delta Lake, a missing field is typically interpreted as `NULL`. The `email IS NOT NULL` condition therefore fails.
- **Why it failed**: This is due to inconsistent schema from the MongoDB source. MongoDB's schemaless nature allows records to omit fields. The `IngestBronze` stage correctly represents this absence as a `NULL` value in the `bronze_customers` table, and the `Scd2Silver` stage propagates this `NULL` to `silver_customers`.
- **Likely failing stage**: The issue originates from the source data. Within the pipeline, the `IngestBronze` stage is where the schema inconsistency is first encountered and handled (by inferring `NULL` for missing fields). The `DqEngine` stage correctly identifies the violation.
- **Rule R001: `price_positive` (for `customer_id: C005`)**
- **What failed**: The `price` field for `customer_id: C005` contains a negative value (`-50.0`), violating the `price > 0` condition.
- **Why it failed**: This is a data entry or business logic error at the source system (MongoDB). The `IngestBronze` and `Scd2Silver` stages have correctly ingested and propagated this erroneous value.
- **Likely failing stage**: The issue originates from the source data. The

Provider: gemini

ORGANIZATION
MY_MDB_ORG

PROJECT
DatabrickAgentMetadata

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

- ClusterRVM
  - admin
  - ecommerce
    - customers
  - local

customers

ClusterRVM > ecommerce > customers

Documents 7 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or Generate query

ADD DATA | UPDATE | DELETE | EXPORT CODE

```
_id: ObjectId('69aa156bd7a70af0991371b1')
customer_id : "C006"
full_name : "Frank Black"
email : null
price : 42
status : "active"
updated_at : 2026-03-05T23:44:43.160+00:00
```

```
_id: ObjectId('69aac54ed3367ddf521550c9')
customer_id : "C002"
full_name : "Brian Hall"
email : "brian.hall@example.com"
price : 109.9
status : "active"
updated_at : 2026-03-06T12:15:10.654+00:00
```

```
_id: ObjectId('69aac54ed3367ddf521550ca')
customer_id : "C004"
full_name : "Daniel Green"
email : "daniel.green@example.com"
price : 75.25
status : "active"
updated_at : 2026-03-06T12:15:10.654+00:00
```

Search data, notebooks, recents, and more... CTRL + P

workspace

Catalog

Serverless Starter Warehouse Serverless 2XS

Type to search...

For you All

- My organization
  - workspace
    - agent
    - agentic_poc
      - bronze_customers
      - dq_issues_log
      - dq_rules
      - metadata_config
      - silver_customers
    - custom
    - default
    - global_temp_like
    - information_schema
    - lf_demo
    - new_default
    - pii_demo
  - system
- Delta Shares Received
  - samples

Catalog Explorer › workspace › agentic_poc ›

metadata_config

Open in a dashboard | Share

Overview | Sample Data | Details | Permissions | Policies | History | Lineage | Insights | Quality

AI Suggested Description

The table stores configuration metadata for various data sources and their corresponding target tables. It includes details such as the name and format of the source, additional options for data processing, and the active status of each configuration. This information is useful for understanding data integration setups, tracking updates, and managing data flow between systems.

Accept | Edit | Send feedback

Filter columns... | AI generate

Column	Type	Comment	Tags	Column masking
source_name	string			
source_format	string			
source_options: keys: "string" values: "string"	map			
target_table	string			
is_active	boolean			
updated_at	timestamp			

About this table

Owner: Rafa Vera

Type: Managed

Data source: Delta

Popularity

Tags

Add tags

Row filter

Add filter

Policies

New policy

Insights

Top users

Rafa Vera

No tables joined in past 30 days

Related assets

[dev jr_vera_ma] poc_pipeline_job

Catalog Explorer › workspace › agentic_poc ›

metadata_config

Open in a dashboard | Share | Create

Overview | **Sample Data** | Details | Permissions | Policies | History | Lineage | Insights | Quality

Ask your question about the sample data... Preview

Reset

Which source names are currently active? | How many different source formats exist? | What are the latest update timestamps per target table?

Sample

	source_name	source_format	source_options	target_table	is_active	updated_at
1	customers_mongo	mongo	{"uri":"mongodb+srv://DBuser [redacted] ym.mongodb.net/?appName [redacted] yWrites=true&w...	workspace.agentic_poc.bronze_custom...	true	2026-03-05T23:48:46.088+00

Catalog

Serverless Starter Warehouse Serverless 2XS

Type to search...

For you All

- My organization
  - workspace
    - agent
    - agentic_poc
      - bronze_customers
      - dq_issues_log
      - dq_rules
      - metadata_config
      - silver_customers
    - custom
    - default
    - global_temp_like
    - information_schema
    - lf_demo
    - new_default
    - pii_demo
  - system
- Delta Shares Received
  - samples

Catalog Explorer › workspace › agentic_poc ›

dq_rules

Open in a dashboard | Share

Overview | Sample Data | Details | Permissions | Policies | History | Lineage | Insights | Quality

AI Suggested Description

The table contains data related to data quality (DQ) rules. It includes information such as the rule ID, rule name, SQL expression used for the rule, severity of the rule, its active status, and the last updated timestamp. This data can be used to manage and monitor data quality checks, evaluate the effectiveness of existing rules, and ensure compliance with data standards.

Accept | Edit | Send feedback

Filter columns... | AI generate

Column	Type	Comment	Tags	Column masking
rule_id	string			
rule_name	string			
rule_sql	string			
severity	string			
is_active	boolean			
updated_at	timestamp			

About this table

Owner	Rafa Vera
Type	Managed
Data source	Delta
Popularity	
Last updated	hace 18 horas
Size	1.8KiB, 1 file

Tags

Add tags

Row filter

Add filter

Policies

New policy

Insights

Top users

Rafa Vera

Catalog Explorer › workspace › agentic_poc ›

bronze_customers

Open in a dashboard | Share

Overview | Sample Data | Details | Permissions | Policies | History | Lineage | Insights | Quality

AI Suggested Description

The table contains information about customers, including their IDs, names, emails, pricing details, and current status. This data can be used for customer relationship management, tracking updates to customer records, and analyzing customer behavior based on pricing and status.

Accept | Edit | Send feedback

Filter columns... | AI generate

Column	Type	Comment	Tags	Column masking
customer_id	string			
full_name	string			
email	string			
price	double			
status	string			
updated_at	timestamp			
source_document_id	string			
ingested_at	timestamp			

About this table

Owner	Rafa Vera
Type	Managed
Data source	Delta
Popularity	
Last updated	hace 18 horas
Size	5.3KiB, 2 files

Tags

Add tags

Row filter

Add filter

Policies

New policy

Insights

Top users

Rafa Vera

No tables joined in past 30 days

Catalog Explorer › workspace › agentic_poc ›

bronze_customers

Overview | **Sample Data** | Details | Permissions | Policies | History | Lineage | Insights | Quality

Ask your question about the sample data...

What are the most common statuses? | Which customers have the highest prices? | How many unique emails are there?

Sample

	customer_id	full_name	email	price	status	updated_at	source_document_id	ingested_at
1	C001	Alice Walker	alice.walker@example.com	125.5	active	2026-03-05T20:33:14.510+00:...	69a9e88a109933b98868844a	2026-03-05T23:49:33.967+00:...
2	C002	Brian Hall	brian.hall@example.com	89.9	active	2026-03-05T20:33:14.510+00:...	69a9e88a109933b98868844b	2026-03-05T23:49:33.967+00:...
3	C003	Carla Stone	carla.stone@example.com	199.0	active	2026-03-05T20:33:14.510+00:...	69a9e88a109933b98868844c	2026-03-05T23:49:33.967+00:...
4	C005	Evelyn Young	evelyn.young@example.co...	-50.0	active	2026-03-05T23:44:43.160+00:...	69aa156bd7a70af0991371b0	2026-03-05T23:49:33.967+00:...
5	C006	Frank Black	null	42.0	active	2026-03-05T23:44:43.160+00:...	69aa156bd7a70af0991371b1	2026-03-05T23:49:33.967+00:...
6	C001	Alice Walker	alice.walker@example.com	125.5	active	2026-03-05T20:33:14.510+00:...	69a9e88a109933b98868844a	2026-03-05T23:10:54.867+00:...
7	C002	Brian Hall	brian.hall@example.com	89.9	active	2026-03-05T20:33:14.510+00:...	69a9e88a109933b98868844b	2026-03-05T23:10:54.867+00:...
8	C003	Carla Stone	carla.stone@example.com	199.0	active	2026-03-05T20:33:14.510+00:...	69a9e88a109933b98868844c	2026-03-05T23:10:54.867+00:...

Table

g Explorer › workspace › agentic_poc ›

silver_customers

Overview | Sample Data | Details | Permissions | Policies | History | Lineage | Insights | Quality

AI Suggested Description

The table contains information about customers, including their identification details and pricing information. It can be used for customer management purposes, tracking customer status and history, and analyzing customer-related metrics. Key attributes include customer ID, full name, email, and pricing details, along with timestamps that indicate when the data was last updated.

Accept | Edit

Send feed

Filter columns...

AI gene

Column	Type	Comment	Tags	Column masking
customer_id	string			
full_name	string			
email	string			
price	double			
status	string			
source_updated_at	timestamp			
updated_at	timestamp			
is_current	boolean			
valid_from	timestamp			
valid_to	timestamp			
record_hash	string			

Catalog Explorer › workspace › agentic_poc ›

dq_issues_log

Open i

Overview | Sample Data | Details | Permissions | Policies | History | Lineage | Insights | Quality

AI Suggested Description

The table logs issues related to data quality rules. It includes details on each identified issue such as the severity, the specific failed condition, and the timestamps of when these issues were detected. This data can be useful for monitoring data quality processes, troubleshooting problems, and improving data governance by providing insights into recurring issues across different rule implementations.

Accept | Edit | Send feedback

Filter columns... | AI generate

Column	Type	Comment	Tags	Column masking
issue_id	string			
rule_id	string			
rule_name	string			
severity	string			
failed_condition	string			
record_key	string			
record_payload	string			
timestamp	timestamp			
pipeline_step	string			

dq_issues_log

Open in a dashboard | Share

Overview | **Sample Data** | Details | Permissions | Policies | History | Lineage | Insights | Quality

Ask your question about the sample data...

Preview

What are the most frequent rule violations? | How many issues per severity level? | When did the most issues occur?

Sample

	issue_id	rule_id	rule_name	severity	failed_condition	record_key	record_payload	timestamp	pipeline_step
1	aab332ef-d7ce-4c38-aa21-b7ea6472403d	R001	price_positive	ERROR	price > 0	C005	{"customer_id":"C005","full_name":"Evelyn Young","email":"evelyn.young@example.com","price":-50.0,"status":"active","source_update...	2026-03-05T23:50:33.035+00:...	DqEngine
2	1bd0a395-4c8e-4855-a996-aa8809ad11...	R001	price_positive	ERROR	price > 0	C005	{"customer_id":"C005","full_name":"Evelyn Young","email":"evelyn.young@example.com","price":-50.0,"status":"active","source_update...	2026-03-05T23:50:59.018+00:...	DqEngine
3	e419ea17-716c-495e-b096-0f0eda95e125	R002	email_not_null	ERROR	email IS NOT NULL	C006	{"customer_id":"C006","full_name":"Frank Black","price":42.0,"status":"active","source_updated_at":"2026-03-05T23:44:43.160Z","update...	2026-03-05T23:50:33.035+00:...	DqEngine
4	d7db1c35-1696-4405-af82-05bbfc31dba6	R002	email_not_null	ERROR	email IS NOT NULL	C006	{"customer_id":"C006","full_name":"Frank Black","price":42.0,"status":"active","source_updated_at":"2026-03-05T23:44:43.160Z","update...	2026-03-05T23:50:59.018+00:...	DqEngine